YIMIAN LIU

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Software Engineer | Agentic AI & Machine Learning | Cornell GPA: 3.95

SUMMARY

Full-stack engineer (TypeScript/React + Python) who has shipped two production AI agent systems — one submitting $1.1M in daily energy-market bids, one used daily by 82% of employees — and owns the quality loop end to end: harness design, evals as release gates, failure analysis from production logs, and cost/latency budgets. Builds with coding agents daily (Claude Code, Cursor) and ships to production in the first week of a new codebase.

EXPERIENCE

Guzman Energy — Senior Software Engineer (Internal title: ORA) Denver, CO · Jan 2026 – Present

- Built and owned an agentic trading system that submits $1.1M bids per day to the SPP market: real-time data ingestion, agentic generation of ML forecast models, trader-in-the-loop UI/UX, LLM-as-a-Judge evals, and human/market feedback — delivering +$842K in incremental P&L and 99.99% on-time acceptance over 7 months.
- Owned the company's self-serve AI analysis and automation layer: a unified MCP server connecting 55 data sources and serving 130K tool calls/month, with fine-grained access control, OAuth2, and audit logging; hosted company-wide AI office hours, and grew daily active users from 0 to 82% of employees.
- Built the eval suites that gate every MCP deploy: 54 regression and 76 capability tasks drawn from real user questions and incidents, with 494 answer checks and an LLM-as-a-Judge scoring each trajectory; hill-climbed the capability pass rate from 36% to 87% in 4 months by filling data gaps, tuning context and prompts, and curating tools, while cutting cost per task by 68%.
- Built an agentic auto-research framework for load/generation forecast: agents propose hypotheses, run experiments, accumulate findings and facts into a shared knowledge graph, and publish ML models as immutable Docker images; an LLM judge rejects leaky runs via trajectory checks, while production output is gated on realized performance — MAE 17% below the best vendor.
- Led and mentored two other engineers to achieve 6x conventional engineering productivity by building an AI-native, self-driven, research-to-production culture: encouraging end-to-end ownership, close end-user engagement, engineer-led innovation; hosting weekly AI research seminars; and automating code review, testing, and CI/CD.
- Grew MCP events 2.8x MoM for four months by using AI agents to identify user pain points from audit logs, rapidly resolving user blockers, leading cross-department onboarding, and hosting biweekly company-wide AI Office Hours.
- Initiated and delivered multiple AI agentic systems, notably: a security scanner that found 6 critical vulnerabilities, including one in a third-party vendor system; a data anomaly and human-error detection system that helped traders proactively avoid $20K+ in losses; an RFP finder that helped Commercial surface a real opportunity in its first month.

Software Engineer Oct 2024 – Dec 2025

- Architected and built from scratch Guzman's internal AI and data platform (Python on GCP, React/TypeScript, BigQuery, SSO), using Cursor and Claude Code intensively: built 16 production apps in 15 months covering trading, operations, accounting, and commercial; owned 3.5B+ rows of BigQuery data; used daily by 74+ employees.
- Built a production RAG system as the company's knowledge base, combining hybrid (BM25 + vector) retrieval, metadata-aware filtering, cross-encoder reranking, and citation-grounded generation; improved Recall@10 from 0.35 to 0.96 and MRR@10 from 0.25 to 0.78.

Elseware Capital — Full Stack Developer (Remote) Jan 2024 – Sep 2024

- Built from scratch a full-stack real estate trading web app with React, FastAPI, and MySQL, supporting asset tokenization, KYC, portfolio, and live blockchain transactions; as the sole engineer on the team, built 95% of the codebase with Claude 3.5 via self-built scaffolds for progressive LLM context management.

Guzman Energy — Frontend Developer Intern (Part-time, Remote) Jul 2023 – Dec 2023

- Built a cross-platform AI scribing mobile app in React Native, using OpenAI Whisper for real-time audio transcription and GPT-3.5 for LLM summarization; optimized WebSocket streaming and Redux caching, cutting transcription latency by 60% and UI render time by 80%.

SELECTED PROJECTS

- **Machine Learning** [Demo] Trained a Transformer model for predator–prey simulation using supervised pretraining and PPO fine-tuning; distilled the model size by 40% for <10ms in-browser inference.
- **Self-Hosted LLM** [Code] Deployed GLM-5.3 (753B MoE) on 4× H200 GPUs with W4A8 quantization; tuned parallelism (TP4, EP), prefix caching, and MTP speculative decoding, achieving 387 tok/s at $0.11/1M tokens (69% below API price).
- **Distributed Systems** [Code] Implemented a sharded Java key/value store using Multi-Paxos and two-phase commit for strongly consistent cross-shard transactions.

TECHNICAL SKILLS

Tools & Languages: Claude Code, Python, TypeScript, React, Node.js, SQL, REST, OAuth, RBAC, OTel, GCP, AWS, K8s
LLM Systems: Claude/OpenAI APIs, tool & MCP, RAG, evals, context management, sandboxed execution, LoRA fine-tuning

EDUCATION

Cornell University — M.Eng., Computer Engineering · GPA: 3.95/4.00 Aug 2022 – Dec 2023
University of Liverpool — B.Eng. (Honors), Electrical & Electronic Engineering · GPA: 3.97/4.00 Sep 2017 – Jul 2021